

NWS Warning Performance Before and After 2025 Staffing Reductions

Craig Rudman

MSDS692: Data Science Practicum I

Anderson College of Business and Computing

Regis University, Denver, CO, USA

crudman@regis.edu

Abstract

This study examines whether warning performance changed following widely reported staffing reductions at the National Oceanic and Atmospheric Administration (NOAA) and the National Weather Service (NWS) beginning in February 2025. Using publicly available verification data from the Iowa Environmental Mesonet (IEM) Cow application, we compare probability of detection (POD), false alarm ratio (FAR), and lead time in advance (LTA) across 116 of the agency's 122 Weather Forecast Offices (WFOs) between a baseline period of 2016 to 2024 and the post-treatment year of 2025. A statistically significant change in performance would be consistent with the hypothesis that staffing reductions affected warning quality and warrant further study. The absence of such a signal, however, would not be definitive, but may provide insight into characteristics of the publicly-available data. In either event, the study cannot establish causality, but contributes a documented, reproducible methodology for evaluating NWS warning performance over time using publicly available data.

I. INTRODUCTION

The National Weather Service (NWS) operates 122 Weather Forecast Offices (WFOs) that collectively warn every county in the United States, issuing roughly 50,000 warnings a year for the events that drive a large share of the nation's weather-related deaths and economic losses [1], [2]. Tornadoes, severe thunderstorms, and flash floods are three such types of events where the margin of safety is determined by the time between a forecaster's warning and the storm's arrival. In one study, advance warning of tornadoes with modest lead times was demonstrated to reduce injury and fatalities by over 40% [3].

In early 2025 the NWS was subject to budget cuts and staff reductions. Beginning February 27, 2025, a government-wide push to shrink the federal workforce cut staff across the NWS through probationary terminations, deferred resignations, and buyouts [4]. By March, the agency's headcount had fallen below 4,000 for the first time in its modern history, roughly 20 percent smaller than at the start of the year, with about 19 percent of forecaster positions vacant and 55 of 122 offices at or above a 20 percent vacancy rate [5]. Former agency leaders and outside scientists warned publicly that degraded staffing would degrade warnings, and in some cases cost lives [5], [6].

This practicum asks whether that concern is realized and reflected in the agency's own output. Using a decade of public warning and storm-report data and the NWS's standard quality metrics, it tests whether 2025 departs from the season-adjusted trend established over the 2016 to 2024 baseline. The design is deliberately narrow and honest about what it can show: it can establish whether a metric changed, coincident with the staffing timeline, but it cannot by itself establish that staffing was the cause.

II. PROBLEM STATEMENT

The research question that this study is designed to address is, "Did NWS staffing cuts in 2025 produce statistically significant changes in warning performance, beyond what normal year-to-year variation produces?"

The question is addressed with nine separate tests: three convective phenomena, tornado (TO), severe thunderstorm (SV), and flash flood (FF), crossed with three performance metrics, Probability of Detection

(POD), False Alarm Ratio (FAR), and Lead Time in Advance (LTA). The phenomena are analyzed separately and never pooled, because they differ in base rate and detection difficulty. Each metric is its own model.

The quantity of interest is not a flat baseline-versus-treatment gap. It is the departure of 2025 (i.e. the treatment) from the baseline trend, adjusted for seasonality and for any drift the metrics were already exhibiting from 2016 to 2024. A metric that was already sliding for nine years should not be attributed to a 2025 intervention; only movement beyond that established trajectory is evidence of a year-10 change.

Three constraints bound the claim from the outset. First, the design is observational; that is, it is a before/after comparison against the public record, not a controlled experiment. Second, no reliable per-office staffing dataset is available publicly, so the analysis cannot attach a 2025 change to staffing at the office level; it treats the cuts as a system-wide event and supports an associational finding only. Third, Local Storm Reports (LSRs), the ground truth for detection, are filed voluntarily, so detection metrics describe performance among reported events, not against the full population of storms. Any change in storm reporting behavior would be a confound for which we would need to account.

III. LITERATURE REVIEW

A small body of work provides a sufficient grounding in forecasting and warning methods. Brooks and Correia provide important insights about the sensitivity of metrics to procedural and methodological changes [7]. Brotzge et al. detail the end-to-end process for detecting and warning tornadoes and identify several challenges such systems encounter [8]. Gourley et al. describe guidance and methods used by the NWS to predict and warn flash floods [9]. The polygon-based verification logic implemented by the data source for severe convective events (i.e. thunderstorms, tornadoes) is documented by Stumpf and Gerard [10] and Stumpf and Stough [11]. Simmons and Sutter connect lead time and false-alarm ratio to casualties, establishing that FAR shifts carry public-safety weight independent of lead time, and that the warning-casualty relationship is empirically complex rather than monotonic [3].

The treatment timeline is established from contemporaneous journalism with editorial oversight and named sources, official transcripts, and government audit material. Reporting in the Washington Post, Scott Dance chronicles the February 27 terminations [4] and the subsequent buyout wave and reduction in headcount [5]. Seth Borenstein of the Associated Press covered the office-level vacancy distribution as of April 4, 2025 [12]. Scott Dance and John Muyskens, again in the Washington Post, reported on overnight-coverage losses at named offices during peak severe season as of May 16, 2025 [13]. On August 7, 2025, Judson Jones wrote in the New York Times of the efforts to fill vacancies starting in June [14]. Hannah Natanson and Brady Dennis provide an update of the staffing gaps going into hurricane season in the Washington Post on September 27, 2025 [15]. Ruby Mellen and Hannah Natanson reported on December 6 in the Washington Post (updated December 8) covering staffing levels going into the winter [16].

IV. METHODOLOGY

A. Data source and provenance

All records come from the Iowa Environmental Mesonet (IEM) COW tool, which ingests NWS warning and LSR products from the operational satellite feed and adds a geospatial verification layer, matching buffered storm reports against warning polygons to compute POD, FAR, and lead time [17]. Because the records originate from the live product stream rather than a post-hoc archive, they reflect what forecasters issued and received in real time. The fetch covered all 122 WFOs for Jan 1, 2016 through June 16, 2026 filtered to TO, SV, and FF, collected in a single run on June 16, 2025. The data is complete with no partial-year bias.

B. Pipeline

A five-stage notebook pipeline (collection, extraction, cleaning, exploratory data analysis, and analysis) writes an immutable checkpoint at each stage, so every figure and statistic is reproducible from the public records. Cleaning casts the logistic outcomes to integer 0/1, imputes null city/county via geocoding, repairs a fixed-width API serialization bug affecting 26 rows, clips non-continental United States offices by office code (leaving 116 WFOs), and derives the temporal design columns. The two analysis-ready tables are:

- **events**: one row per issued warning; 260,537 rows
- **stormreports**: one row per observed event; 376,963 rows.

C. Temporal design

Each study year begins February 27 and runs to the next February 26. Seasons are three-month windows breaking on the 27th of February, May, August, and November, anchored to the February 27 treatment date so every boundary is treatment-relative. The three model terms are derived from the engineered **YYYY-Season** label during cleaning:

- **season_cat**: Spring/Summer/Fall/Winter, categorical
- **study_year**: 1-based index, 1 for 2016 through 10 for 2025 (i.e. the numeric trend term)
- **is_year10**: 0/1, where 1 is set only for 2025

D. The nine models

Every fit shares the right-hand side on all ten years, via the statsmodels formula interface.

```
outcome ~ C(season_cat) + study_year + is_year10
```

The left-hand side (i.e. **outcome**) varies as follows:

- **POD**: logistic regression on storm-report rows; outcome **warned**.
- **FAR**: logistic regression on warning rows; outcome **verify**.
- **LTA**: ordinary least squares (OLS) on warned storm reports only; outcome **leadtime** in minutes.

study_year absorbs baseline drift across years 1 to 9. The **is_year10** coefficient is the year-10 departure from that season-adjusted trend, the quantity of interest. Logistic coefficients are reported as odds ratios, OLS coefficients in minutes. Each fit logs sample size, rows dropped, and the fitted summary; per-**season_cat** counts and group rates are checked for thin cells and separation before fitting.

To guard against false positives, two checks run alongside the nine tests: a placebo test drops year 10 and refits the same structure with the indicator pointed at each interior baseline year in turn; a real effect is credible only if it stands clearly outside the spread of placebo effects from innocent years. A Benjamini-Hochberg false-discovery-rate (FDR) correction adjusts the nine p-values together at $\alpha = 0.05$.

V. DATA ANALYSIS

Four descriptive views characterize the data the nine fits rest on.

A. Geographic distribution

The storm reports cover the continental United States (CONUS) but not evenly: SV and TO concentrate through the plains and southeast while FF spreads more widely, including the mountainous west, so the footprint is national but the density is regional.

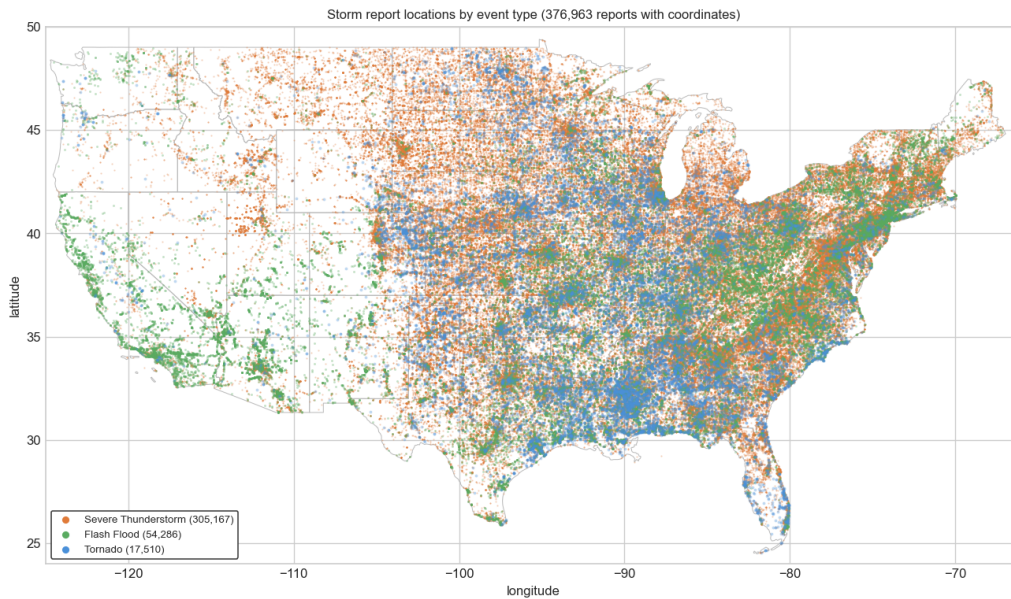


Fig. 1. Geographic distribution of storm reports (2016 to 2025); national footprint, regional density.

B. Workload distribution

Warning volume is shared unevenly across the office network. Of the 116 in-scope offices the busiest five issue about 12% of all warnings and the busiest twenty about 36%, while the entire bottom half together issue only about 22%. Norman, OK (OUN) leads by a wide margin, near double the next office, reflecting the southern plains' severe-weather load. The busiest offices track the geography where storm reports concentrate, though the leaders are phenomenon-specific. OUN heads severe storm and tornado, while flash-flood volume shifts to the desert Southwest.

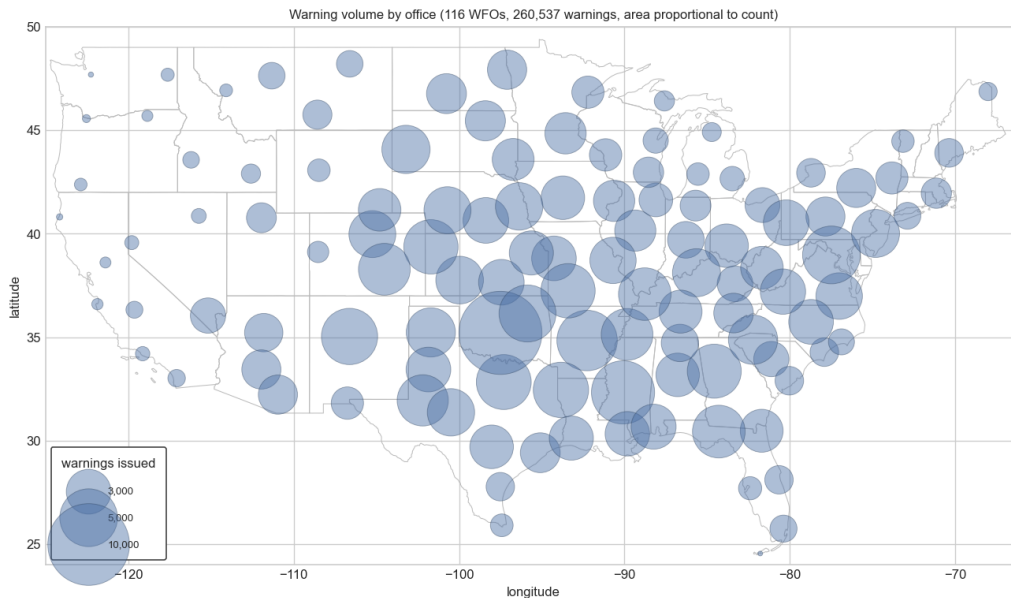


Fig. 2. Warning volume by office; a few high-volume offices dominate the national totals.

C. Yearly volumes

Across the study span, both source tables carry every year with none missing or collapsing, and total volume fluctuates year to year as the weather does. Warnings range from about 21,400 to 31,400, a spread

of roughly 47%, and storm reports from about 32,900 to 44,200, a spread of roughly 34%, with 2025 the highest-volume warning year and among the highest for reports.

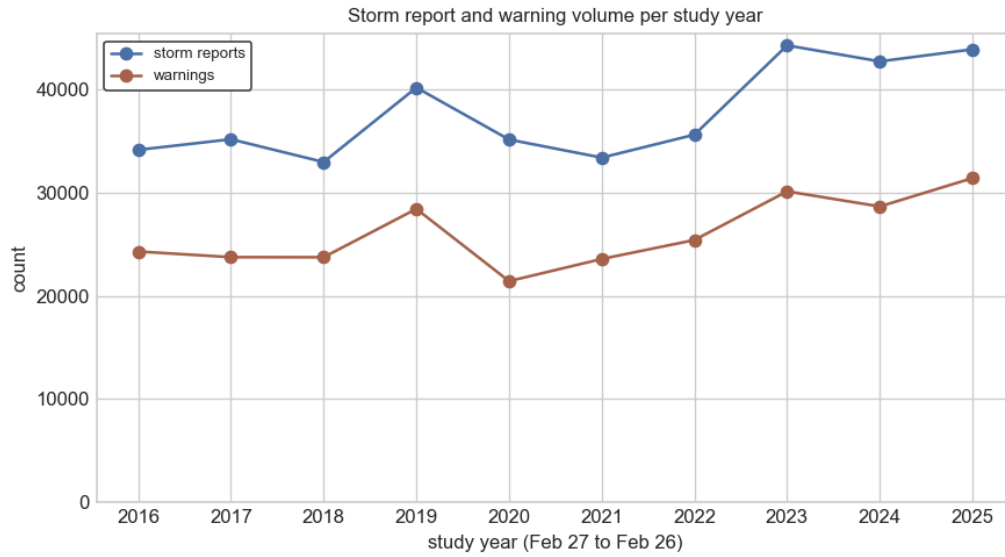


Fig. 3. Storm report and warning volume per study year (2016 to 2025)

D. Seasonal variation

Each event type has a distinct seasonal signature, with tornado and severe-storm reporting peaking in spring and summer and flash-flood reporting concentrating in but spreading past the warm season. This is the within-year structure the $\mathbf{C}(\text{season_cat})$ term absorbs so the study-year and year-10 terms read net of season.

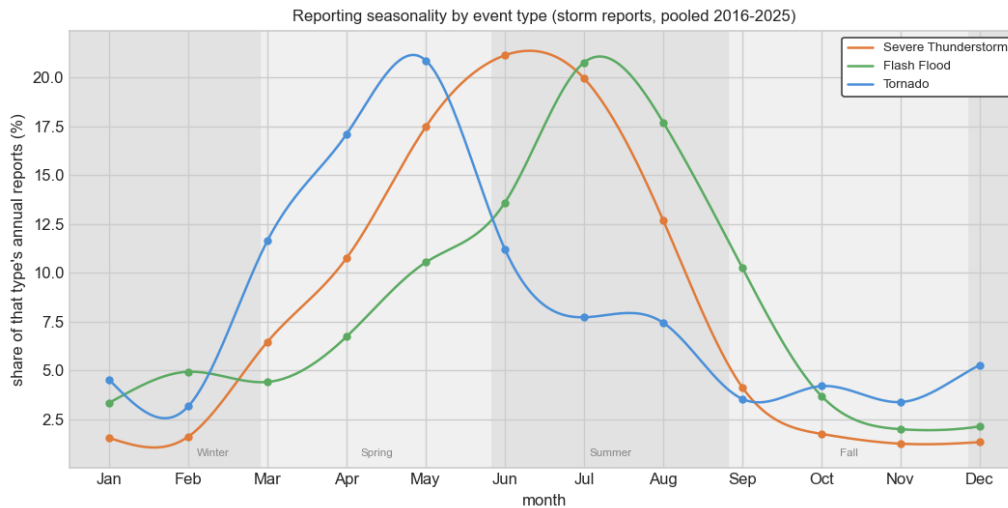


Fig. 4. Each event type has a distinct seasonal signature. The shaded bands mark four engineered seasons; note they break on the 27th of February, May, August, and November (anchored to the treatment anniversary), not on the canonical calendar dates.

Each metric was inspected as a raw yearly rate before modeling, subjected to a baseline-only drift check (years 1 to 9) and the full ten-year fit. Each of the nine fits regresses one metric for one phenomenon, with detection (POD) and false alarms (FAR) fit by logistic regression and lead time (LTA) by ordinary least squares. Every fit carries the same right-hand side, a season term, a baseline-trend term, and a year-10

indicator, so the trend absorbs any drift across years 1 to 9 and the indicator measures how far 2025 departs from where that trend would land. That departure is the quantity of interest, reported below for all nine fits. For the logistic metrics it is an odds ratio, where 1.0 is no change, above 1.0 more likely, and below 1.0 less likely. For lead time it is a difference in minutes, where 0 is no change. Each cell pairs the departure with its p-value, and bold marks the results that survive the false-discovery-rate correction across all nine tests.

TABLE I
YEAR-10 DEPARTURE BY METRIC AND EVENT TYPE

Metric	SV	TO	FF
POD	OR 1.01 ($p = 0.37$)	OR 1.09 ($p = 0.17$)	OR 1.04 ($p = 0.26$)
FAR	OR 1.15 ($p \approx 10^{-17}$)	OR 1.03 ($p = 0.56$)	OR 0.87 ($p \approx 3 \times 10^{-4}$)
LTA	-0.2 min ($p = 0.10$)	-0.6 min ($p = 0.15$)	-13.0 min ($p \approx 2 \times 10^{-13}$)

POD shows no significant year-10 departure for any phenomenon. Three results survive FDR: SV FAR, FF FAR, and FF LTA. The placebo test is the decisive screen, and of the three, only SV FAR clears it. FF FAR and FF LTA produced effects in baseline years as large as their year-10 effect, so the same machinery flags "changes" in innocent years and these cannot be treated as findings.

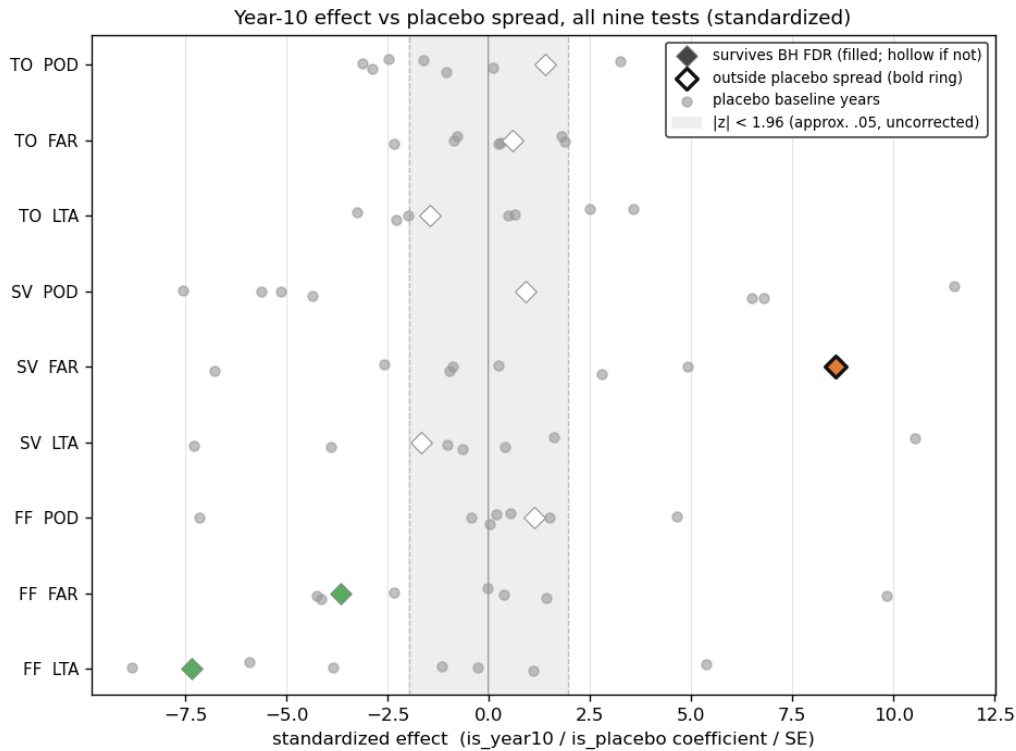


Fig. 5. Year-10 departure (diamond) against the placebo effects from innocent baseline years (grey) for each of the nine fits, on a standardized scale. SV FAR is the lone diamond sitting clearly outside its placebo spread; FF FAR and FF LTA fall among placebo effects as large as their own, which is why they do not survive the screen.

E. The one standout

Severe-thunderstorm false-alarm odds rose about 15 percent in 2025 (95% CI +11.6% to +19.1%, $p < 0.001$), surviving FDR and sitting clearly outside its placebo spread, while SV POD and SV LTA held flat. On the rate scale, the 2025 SV false-alarm rate (0.543) sits about 0.036 above where the season-adjusted baseline trend lands (0.507).

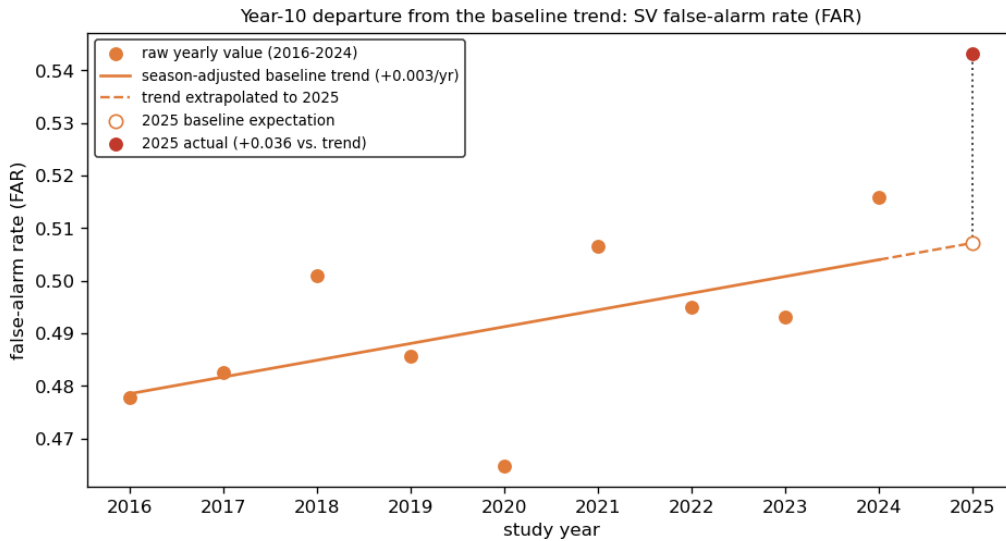


Fig. 6. Severe-thunderstorm false-alarm rate, raw yearly values (2016 to 2024) with the season-adjusted baseline trend extended to 2025.

F. Larger warnings as a possible confound

Because the one credible finding is a rise in false alarms, a natural follow-on question is how the warning behavior may differ. One mechanism that would mechanically raise false alarms without any change in skill or staffing would be if forecasters drew larger, more precautionary warning polygons in 2025; if each warning covered more ground, more of it might go unverified. The events table records warning polygon area in square kilometers as **carea**. This was examined directly with the same departure-from-trend structure used for the nine fits using **log(carea)** since polygon area is heavily right-skewed.

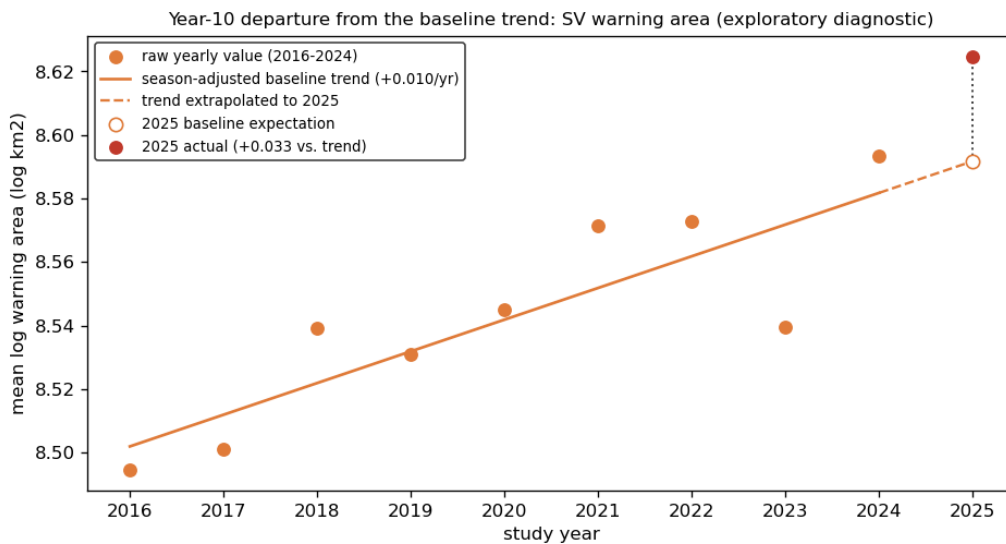


Fig. 7. Severe-thunderstorm warning polygon area on the log scale, raw yearly values (2016 to 2024) with the season-adjusted baseline trend extended to 2025. The 2025 actual sits about 4 percent above the trend's expectation.

Severe-thunderstorm warning area did depart upward in 2025, about 4 percent above where its season-adjusted baseline trend lands (95% CI +2.9% to +5.4%, $p < 0.001$), consistent with larger, more precautionary polygons. This is descriptive context for the false-alarm finding only. It is not a finding. What it establishes is that larger-area warning is a live, unexcluded mechanism for the severe-thunderstorm false-alarm rise, deserving of further examination.

VI. CONCLUSION

The research question was whether the 2025 staffing cuts coincided with changes in NWS warning performance beyond normal year-to-year variation. Across nine tests, three convective phenomena crossed with three metrics, the answer is mostly no, with one exception that survives every screen.

Detection (POD) did not move for any phenomenon, the result that would most directly signal that storms were being missed. Of the nine fits, three departures survived the false-discovery-rate correction, but the placebo test, the decisive screen, reduced those three to one. Flash-flood FAR and flash-flood LTA produced effects in innocent baseline years as large as their 2025 effect, so the same machinery flags "changes" where none was caused; they cannot be read as findings. Only severe-thunderstorm false alarms cleared both screens: a roughly 15 percent rise in false-alarm odds (95% CI +11.6% to +19.1%), standing clearly outside its placebo spread, while severe-thunderstorm detection and lead time held flat.

The design is observational and the staffing event is treated as system-wide, so the result is associational: 2025 severe-thunderstorm false alarms rose coincident with the staffing timeline, not demonstrably because of it. No per-office staffing data was available to tie the change to staffing at the office level, the binding limit on attribution. The detection metrics describe performance among voluntarily reported events, not the full storm population. And the descriptive check on warning area, which found severe-thunderstorm polygons about 4 percent larger than trend in 2025, identifies an alternative, non-staffing mechanism for the very rise this study isolated: larger precautionary polygons mechanically cover more unverified ground. The false-alarm rise is real; its cause is not settled.

What the study establishes is narrow and defensible. One severe-thunderstorm metric departed from its season-adjusted baseline in 2025 in a way that ordinary variation does not explain, and eight others did not. The widely voiced concern that degraded staffing would degrade warnings is neither confirmed nor refuted by this record alone; the one signal present is consistent with that concern but equally consistent with a shift in warning practice. Resolving which would require the extensions outlined in the future work: credible per-office staffing data, a control group to net out broad weather and reporting shifts (difference-in-differences), a load-by-staffing interaction to target where any effect would concentrate, and cluster-robust standard errors throughout. Until then, the appropriate reading is that the agency's own performance record shows one change worth watching, not a system-wide collapse, and that the question of cause remains open.

REFERENCES

- [1] National Weather Service, "National Weather Service: Forecasts and Service," <https://www.weather.gov/about/forecastsandservice>, n.d.
- [2] A. B. Smith and R. W. Katz, "U.S. Billion-dollar Weather and Climate Disasters: Data Sources, Trends, Accuracy and Biases," *Natural Hazards*, vol. 67, no. 2, pp. 387–410, 2013.
- [3] K. M. Simmons and D. Sutter, "Tornado Warnings, Lead Times, and Tornado Casualties: An Empirical Investigation," *Weather and Forecasting*, vol. 22, no. 3, pp. 599–608, 2007.
- [4] S. Dance, "Mass firings hit critical National Weather Service and NOAA offices," *The Washington Post*, Feb. 2025.
- [5] —, "National Weather Service buyouts will leave gaps as storm season ramps up," *The Washington Post*, Apr. 2025. [Online]. Available: <https://www.washingtonpost.com/weather/2025/04/16/national-weather-service-buyouts-staff-shortages-trump-administration/>
- [6] M. Baker, "Former Weather Service Leaders Warn Staffing Cuts Could Lead to 'Loss of Life'," *The New York Times*, May 2025.
- [7] H. E. Brooks and J. Correia, James, "Long-Term Performance Metrics for National Weather Service Tornado Warnings," *Weather and Forecasting*, vol. 33, no. 6, pp. 1601–1614, 2018.
- [8] J. Brotzge and W. Donner, "The Tornado Warning Process: A Review of Current Research, Challenges, and Opportunities," *Bulletin of the American Meteorological Society*, vol. 94, no. 11, pp. 1715–1733, 2013.
- [9] J. J. Gourley *et al.*, "Evaluation of Tools Used for Monitoring and Forecasting Flash Floods in the United States," *Weather and Forecasting*, vol. 27, no. 1, pp. 158–173, 2012.
- [10] G. J. Stumpf and A. E. Gerard, "National Weather Service Severe Weather Warnings as Threats-in-Motion," *Weather and Forecasting*, vol. 36, no. 2, pp. 627–643, 2021.
- [11] G. J. Stumpf and S. M. Stough, "A Geospatial Verification Method for Severe Convective Weather Warnings: Implications for Current and Future Warning Methods," *Weather and Forecasting*, vol. 39, no. 5, pp. 689–710, 2024.
- [12] S. Borenstein, "Nearly half of National Weather Service offices are critically understaffed, experts warn," *PBS NewsHour* (Associated Press), Apr. 2025. [Online]. Available: <https://www.pbs.org/newshour/politics/nearly-half-of-national-weather-service-offices-are-critically-understaffed-experts-warn>
- [13] S. Dance and J. Muyskens, "Where local forecast offices no longer monitor weather around the clock," *The Washington Post*, May 2025. [Online]. Available: <https://www.washingtonpost.com/weather/2025/05/16/weather-service-offices-overnight-cuts-map/>

- [14] J. Jones, “Weather Service Is Hiring Hundreds After Sweeping Cuts Earlier This Year,” The New York Times, Aug. 2025.
- [15] H. Natanson and B. Dennis, “National Weather Service at ‘breaking point’ as storm approaches,” The Washington Post, Sep. 2025. [Online]. Available: <https://www.washingtonpost.com/politics/2025/09/27/national-weather-service-staffing-crisis/>
- [16] R. Mellen and H. Natanson, “Winter is coming. Not all weather offices are ready,” The Washington Post, Dec. 2025. [Online]. Available: <https://www.washingtonpost.com/weather/2025/12/06/winter-national-weather-service-staff-shortages/>
- [17] Iowa Environmental Mesonet, “IEM COW: NWS Storm Based Warning Verification,” <https://mesonet.agron.iastate.edu/cow/>, n.d., data available from June 8, 2005; accessed June 6, 2026.